

When Is an Inverse a Function?

TEACHER KEY | Algebra II Honors | Unit 1 | Day 12 | TEK A2.2.C

Objective: I can determine whether the inverse of a function is a function, choose a domain restriction that makes it one, and apply this to a quadratic whose vertex is not at the origin.

Vocabulary

One-to-one — every output comes from exactly one input. Detected by the **horizontal line test**.

Key fact — the inverse of a function is always a **relation**, but a **function** only when the original is one-to-one.

Domain restriction — deliberately shrinking the domain to force one-to-one behavior. Restricting the domain of f restricts the **range** of f^{-1} .

Part 1 — Why $f(x) = x^2$ Fails

x	-3	-2	-1	0	1	2	3
$f(x) = x^2$	9	4	1	0	1	4	9

Repeated outputs: **1, 4, 9** So f is **not one-to-one**

Swapping produces both (1, 1) and (1, -1). State precisely which test the inverse fails and why:

The **vertical line test** — the input 1 would need two outputs.

Solving $y = x^2$ for x gives $x = \pm\sqrt{y}$, and the $\pm$ is the whole problem.

Part 2 — Restrict and Repair

Restriction on x^2	Half of the parabola kept	Resulting $f^{-1}(x)$	Range of f^{-1}
$x \geq 0$	right half	$+\sqrt{x}$	$[0, \infty)$
$x \leq 0$	left half	$-\sqrt{x}$	$(-\infty, 0]$

Both restrictions work. Is one more correct than the other? **no - $x \geq 0$ is just the convention**

Part 3 — A Vertex That Is Not at the Origin

Now take $g(x) = (x - 2)^2$. The same idea applies, but you must locate the turning point first.

Vertex of g : **(2, 0)** So the natural restriction is **$x \geq 2$ (or $x \leq 2$)**

Now find the inverse. Swap and solve for y , keeping track of the $\pm$.

$$x = (y - 2)^2 \rightarrow \pm\sqrt{x} = y - 2 \rightarrow y = 2 \pm \sqrt{x}$$

With the restriction $x \geq 2$ on g , the inverse is $g^{-1}(x) = 2 + \sqrt{x}$ Domain of g^{-1} **$[0, \infty)$**

Why does the 2 appear OUTSIDE the radical in the inverse when it was inside the square in g ?

Undoing happens in **reverse order**: take the root first, then add 2.

Part 4 — Which Need No Restriction?

Function	One-to-one?	Restriction needed?	Inverse
x^2	no	yes	$\sqrt{x}$
x^3	yes	no	$\sqrt[3]{x}$
2^x	yes	no	$\log_2(x)$
$ x $	no	yes	$\pm x$
$1/x$	yes	no	$1/x$

State the pattern as a rule about increasing and decreasing:

A function that only increases or only decreases is **one-to-one**; one that turns around is not.

Part 5 — Going Further

1. Restrict $h(x) = (x + 3)^2$ so its inverse is a function, then write the inverse.

Restrict **$x \geq -3$** $h^{-1}(x) = -3 + \sqrt{x}$

2. Explain why $1/x$ is its own inverse, using the swap.

3. Is there a restriction of $|x|$ whose inverse is a function? Give one. **$x \geq 0$, giving $f^{-1}(x) = x$**

Exit Ticket

Consider $f(x) = (x - 5)^2$ with no restriction.

Is its inverse a function? **no** Give a restriction that fixes it: **$x \geq 5$ (or $x \leq 5$)**

With that restriction, $f^{-1}(x) = 5 + \sqrt{x}$ Explain in one sentence why the restriction is needed:
